

ELECTRONICS DIY TOOLS

For Hobbyists And Professionals



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Use of proper tools is important for making any project properly. Here are some tools that are in common use by amateurs as well as professionals.

Soldering tools

Soldering tools are needed for performing efficient soldering work. Soldering is an art by which electronic components are connected with each other, so that these can communicate in a circuit and perform a specific task. Electronic components are held together on a board called a printed

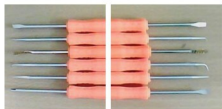


Fig. 1: Tools shown in the two images are actually pairs and form six types of different tools: Knife pair is useful for cutting tracks of PCB, chip hold pair for holding surface mount devices while soldering, cleaning brush pair for removing solder from PCB, hook pair for removing components after de-soldering, reamer pair for transforming holes on PCB to larger-desired size and fork pair for aligning component leads to desired direction

TABLE I
SOLDERING TOOLS

Name of tool	Function
Variable soldering station	Soldering stations have a soldering pen/iron and a base unit with a knob to vary temperature of the soldering iron tip as per needs. Most soldering jobs are performed at 250°C/15W knob setting. For bigger joints, you can use 350°C/25W knob setting, too.
35-watt soldering iron	This soldering iron is part of Max Gold 535 station and you need to use Bevel 4C bit with it for almost all kinds of soldering jobs. Tip of the bit is the area that melts the solder, so it should be properly tinned frequently for best results. Always leave a blob of solder on the tip after you finish soldering.
Tripod PCB holder	It is needed for holding the PCB while performing soldering work. The PCB can be rotated at any angle as per needs.
De-soldering pump	This tool is needed to correct any mistakes that you may have committed while soldering. It helps extract the excess or wrongly-formed solder from the solder joint.
Connections	The light-green area is the copper track that connects two component leads with the help of a molten solder (white, conical objects). The dark-green area is the solder side of the PCB.
Fine 26SWG solder wire	You should use very fine solder wires (of size 26SWG) to avoid excess feeding of solder to the joint. It is better to use 60 per cent tin and 40 per cent lead based solder wire, as lead-free ones need higher melting temperatures and are difficult to inspect for correct joint formation. Use a solder wire with at least five cores of flux within.
Thick 22SWG solder wire	For bigger joints, use a thicker solder wire (of size 22SWG). Composition should be 60/40 (same as fine solder wire).
De-solder braid	De-solder braid, or de-solder wick, is extremely useful to wick away (extract away) excess solder from an incorrect solder joint. Place it between the joint and the tip of the soldering iron and it will magically absorb all the solder from the joint. It has flux built-in that turns to smoke when heated.
Flux paste	Flux is applied to the area where solder joints need to be formed. Action of flux is simple; it is corrosive in nature at soldering temperatures and non-corrosive at normal temperatures. Due to its corrosive nature it removes oxides and other impurities from soldering surfaces while soldering.
PCB cleaner	The PCB cleaner is used to clean the PCB after soldering work to remove left-over flux.
Cleaning sponge	It is first made wet and then squeezed to the level so that water does not drip from it but instead moistens. It is used to clean and tin the tip of the soldering iron. This is a cellulose-type sponge.